Predicting Diabetes Outcome

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Introduction

Question:

- What risk factors best predict the development of diabetes?

Background:

- ~37.3 million people in the US have diabetes
- Type 2 Diabetes is the most common form
- Diabetes can cause coronary artery disease, heart attack, stroke, and other complications
- Preventing and management of diabetes is crucial for a healthy lifestyle

Data Collection

- Who?
  - Data was collected by a Kaggle
- What?
  - “A Comprehensive Dataset for Predicting Diabetes with Medical & Demographic Data”
  - Individual human level data (not longitudinal)
  - Each row is a different person
  - Variables related to development of diabetes -> specific variables listed on the next page!
- When?
  - Updated 3 years ago
- Where?
  - Collected from various EHRs
  - EHRs are digital patient medical records, and are collected by hospitals and clinics whenever a patient is there

Variable Descriptions

diabetes (response)	1 indicates the presence of diabetes and 0 indicates absence of diabetes.
gender	The biological sex of the individual. Three categories: male, female, and other.
age	The age of individual. Age ranges from 0-80 in our dataset.
hypertension	A medical condition in which blood pressure in the arteries is persistently elevated. 0 indicates no hypertension and 1 indicates hypertension.
heart_disease	0 Indicates no heart disease, 1 indicates heart disease.
smoking_history	Categorized in 4 ways: never smoked, former smoker, current smoker, or no info.
bmi	A measure of body fat based on weight and height. BMI less than 18.5 is classified as underweight, 18.5-24.9 as normal, 25-29.9 as overweight, and 30 or more as obese. Here, we consider BMI as a continuous variable, not broken up by level.
HbA1c_level	Hemoglobin A1c Levels is a measure of a person's average blood sugar level over the past 2-3 months.
blood_glucose_level	The amount of glucose in the bloodstream at a given time.

Manipulations and Potential Issues

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Data Manipulations

- Variables hypertension, heart disease, smoking history, gender, and diabetes were turned into a factor variable
- Levels for smoking history were condensed into four categories rather than six

Potential Issues

- Data does not specify type 1 or type 2 diabetes
  - T1D is an autoimmune disease and cannot be prevented
  - T2D develops later in life and can often be prevented
- Smoking history variable
  - Original categories overlapped with the 6 categories
  - Had to specify interpretations of the "no info"/ "no history" category
- Kaggle data

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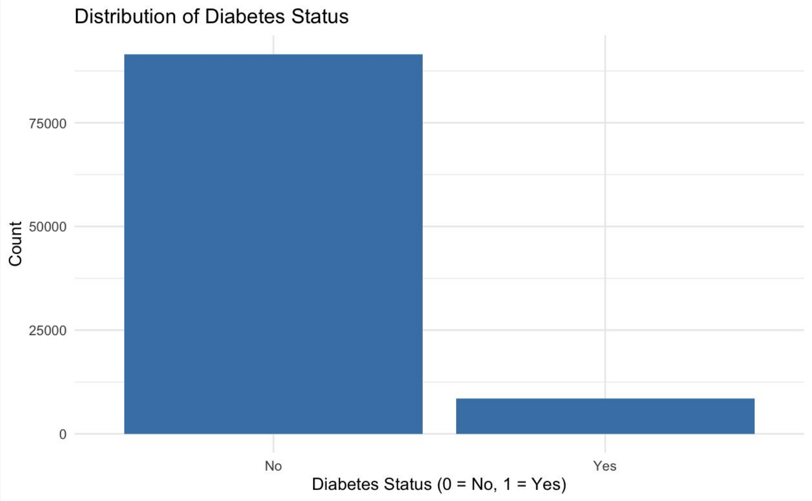
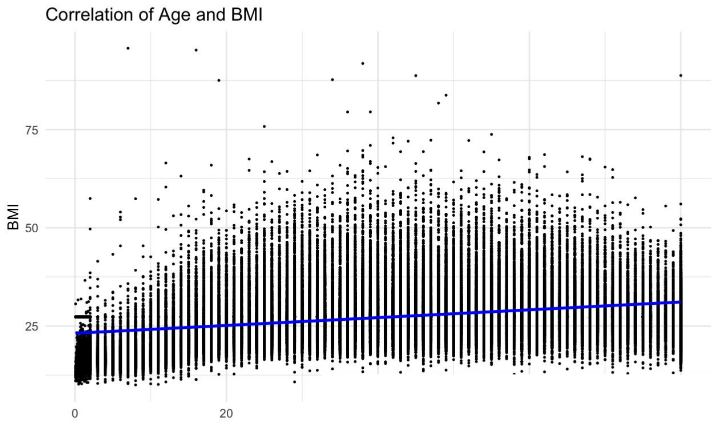
Our Response Variable - Presence of Diabetes

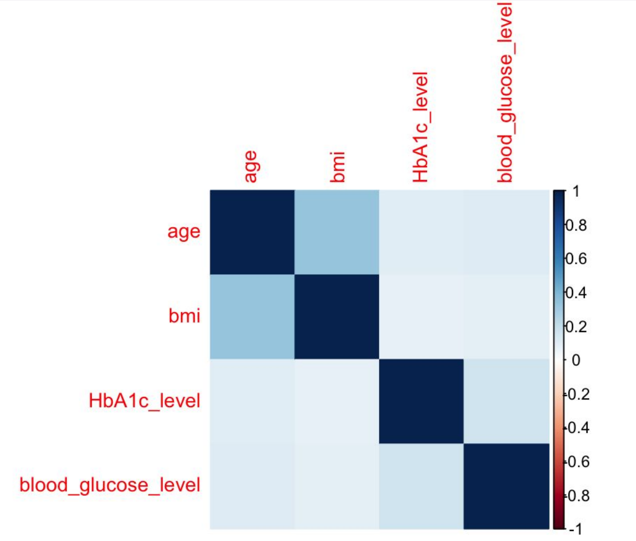
- Main Variable: Presence or absence of diabetes
- Binary variable (0 = failure/ no diabetes, 1 = success/ diabetes)
- Histogram shows imbalance of successes and failures
  - 8.5% of individuals have diabetes, while 91.5% do not



EDA- Development of Diabetes

- Greatest correlation between quantitative variables is age and bmi
- Correlation is low: 0.3373958

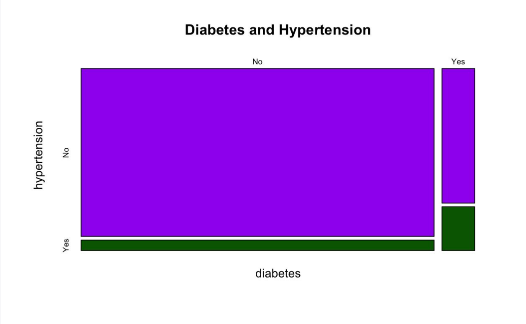


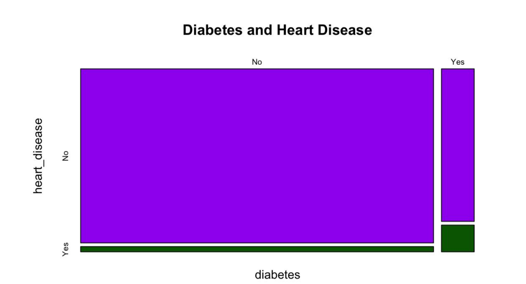


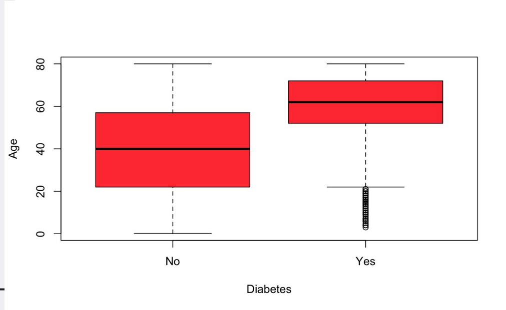
	Age	BMI	HbA1c level	Blood Glucose Level
Age	1.0000000	0.33739578	0.10135366	0.1106723
BMI	0.3373958	1.00000000	0.08299716	0.0912614
HbA1c level	0.1013537	0.08299716	1.00000000	0.1667329
Blood Glucose Level	0.1106723	0.09126140	0.16673293	

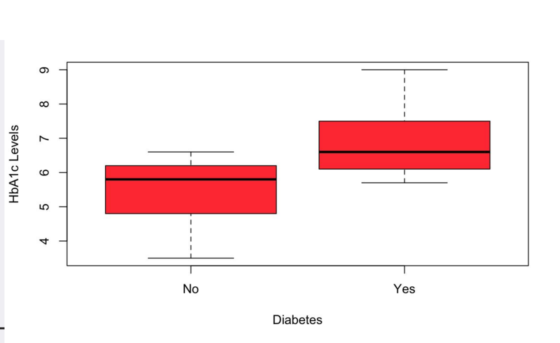
EDA Cont.

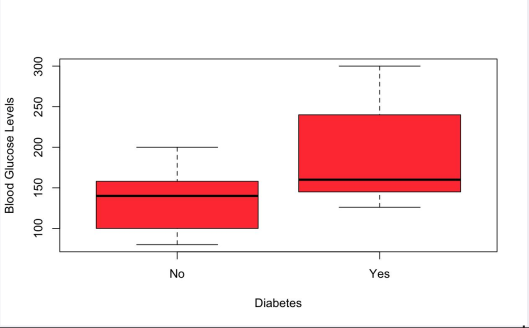
- Individuals with diabetes tend to have substantially higher HbA1c and blood glucose levels compared to those without diabetes, suggesting a strong positive association.
- Presence of hypertension and heart disease also seem to be positively correlated correlated











Logistic Regression

Logistic Regression:

- Used to predict the likelihood of an event occurring
- Explores the association between a binary outcome and one or more risk factors

Our data:

- Diabetes variable is suitable to be a logistic regression response variable because it is a binary outcome (0 indicates no diabetes, 1 indicates having diabetes)

Model

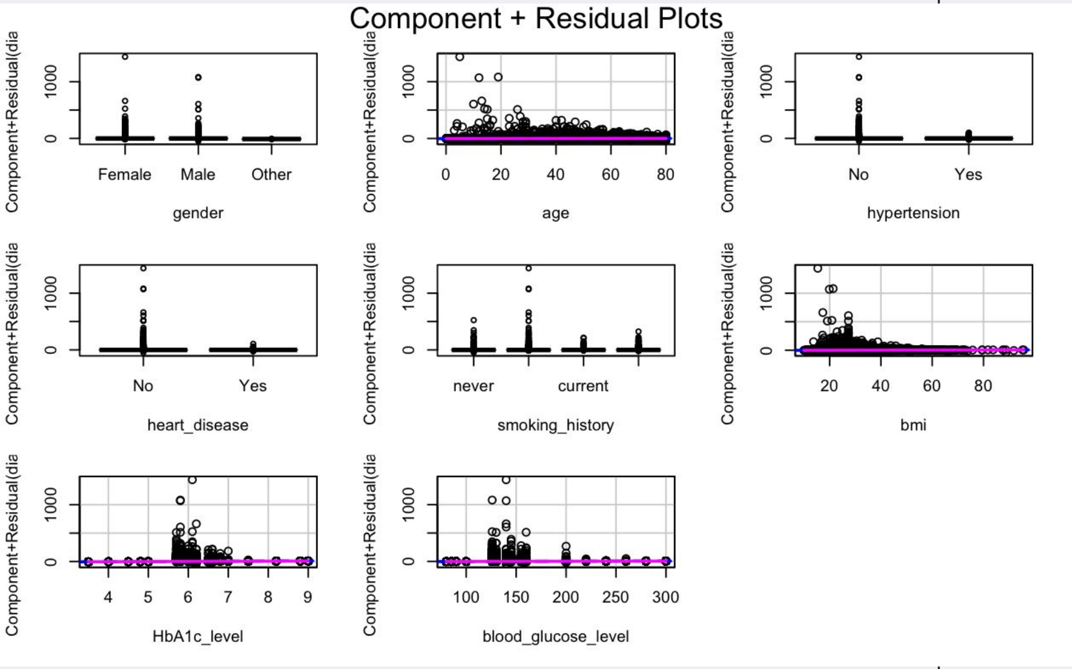
$\log(p/1-p) = \beta 0 + 0.274(\text{Male}) + 0.0462(\text{age}) + 0.744(\text{hypertension}) + 0.739(\text{heart\_disease}) - 0.574(\text{No Info}) + 0.156(\text{current}) + 0.031(\text{former}) + 0.089(\text{bmi}) + 2.34(\text{HbA1c\_level}) + 0.033(\text{blood\_glucose\_level})$

Characteristic	OR	95% CI	p-value
gender			
Female	—	—	
Male	1.32	1.23, 1.41	<0.001
Other	0.00		>0.9
age	1.05	1.04, 1.05	<0.001
hypertension			
No	—	—	
Yes	2.10	1.92, 2.31	<0.001
heart_disease			
No	—	—	
Yes	2.09	1.86, 2.36	<0.001
smoking_history			
never	—	—	
No Info	0.56	0.51, 0.62	<0.001
current	1.17	1.04, 1.32	0.010
former	1.03	0.95, 1.12	0.5
bmi	1.09	1.09, 1.10	<0.001
HbA1c_level	10.4	9.69, 11.1	<0.001
blood_glucose_level	1.03	1.03, 1.03	<0.001

Abbreviations: CI = Confidence Interval, OR = Odds Ratio

Assumptions

- Binary Outcome ✔
  - Diabetes variable is coded as 1 = diabetes or 0 = no diabetes
- Independence ✔
  - Each individual only has one set of measurements
- Linearity ✔
  - Magenta line is falling on top of our dashed blue line with no curvature
  - Suggest linear relationship between outcome and predictors



Assumptions cont.

- Multicollinearity ✔
  - All of our VIFs are clearly close to 1, so we have no major concerns

	GVIF	Df	GVIF^(1/(2*Df))
gender	1.031735	2	1.007841
age	1.145628	1	1.070340
hypertension	1.049174	1	1.024292
heart_disease	1.072262	1	1.035501
smoking_history	1.066301	3	1.010757
bmi	1.047582	1	1.023514
HbA1c_level	1.051067	1	1.025215
blood_glucose_level	1.056526	1	1.027875

- Sufficient Sample Size ✔
  - Total sample size is 100,000, which is sufficiently large for our logistic regression with 8 predictors
  - We also have enough observations within each level: 8500/15 = 567 maximum predictors in our model, and we have 8

Model and Measures of Association

log(p/1-p)= β0 + 0.274(Male)+ 0.0462(age) + 0.744(hypertension) + 0.739(heart _disease) - 0.574(No Info) +0.156(current) + 0.031(former) + 0.089(bmi) + 2.34(HbA1c_level) + 0.033(blood_glucose_level)

```
    Null deviance: 58163  on 99999  degrees of freedom
Residual deviance: 22630  on 99988  degrees of freedom
AIC: 22654
```

Coefficients

```
Call:
glm(formula = diabetes ~ ., family = binomial(), data = diab)

Coefficients:
                          Estimate Std. Error z value Pr(>|z|)    
(Intercept)             -27.241956   0.290674 -93.720  < 2e-16 ***
genderMale                0.274409   0.036091   7.603 2.89e-14 ***
genderOther              -9.527627 102.805629  -0.093  0.92616    
age                       0.046182   0.001121  41.183  < 2e-16 ***
hypertensionYes           0.744161   0.047065  15.811  < 2e-16 ***
heart_diseaseYes          0.738888   0.060662  12.180  < 2e-16 ***
smoking_historyNo Info   -0.573708   0.048881 -11.737  < 2e-16 ***
smoking_historycurrent    0.156286   0.060573   2.580  0.00988 ** 
smoking_historyformer     0.030716   0.043782   0.702  0.48295    
bmi                       0.088989   0.002553  34.854  < 2e-16 ***
HbA1c_level               2.340200   0.035779  65.407  < 2e-16 ***
blood_glucose_level       0.033355   0.000482  69.197  < 2e-16 ***
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Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for binomial family taken to be 1)
```

AIC

Likelihood Ratio Test

```
lltest<-diabMod1$null.deviance-diabMod1$deviance
llpvalue<-1-pchisq(lltest,6)
llpvalue
```

```
[1] 0
```

Significant Variables and Interpretations

Gender:
While adjusting for other variables in the model, the odds of having diabetes for males are 1.32 times higher than the odds of having diabetes for females.

Age:
While adjusting for other variables in the model, as age increases by one year, the odds of having diabetes increases by a factor of 1.05.

Hypertension:
While adjusting for other variables in the model, the odds of having diabetes for an individual with hypertension are 2.1 times higher than the odds of having diabetes for individuals without hypertension.

Heart Disease:
While adjusting for other variables in the model, the odds of having diabetes for an individual with heart disease are 2.09 times higher than the odds of having diabetes for individuals without heart disease.

Significant Variables and Interpretations

Smoking History:
While adjusting for other variables in the model, the odds of having diabetes for an individual who is currently smoking are 1.17 times higher than the odds of having diabetes for individuals who have never smoked.

BMI:
While adjusting for other variables in the model, as BMI increases by one unit, the odds of having diabetes increases by a factor of 1.09.

HbA1c Level:
While adjusting for other variables in the model, as HbAIc level increases by one unit, the odds of having diagnosis increases by a factor of 10.4.

Blood Glucose Level:
While adjusting for other variables in the model, as blood glucose increases by one unit, the odds of having diabetes diagnosis increases by a factor of 1.03.

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Limitations and Future Research

Limitations

- Specifications of types of diabetes
- More accurate collection of data and better
- “No info” category in the smoking history variable causes confusion in interpretations

Future Research

- Additional variables to help predict incidence of diabetes
- Follow-up of individuals who have diabetes and management factors
- Using treatments as variables to see decrease in blood glucose levels

References

"Diabetes: What It Is, Causes, Symptoms, Treatment & Types." Cleveland Clinic, 17 Feb. 2023, my.clevelandclinic.org/health/diseases/7104-diabetes.

Mustafa, Mohammed. "Diabetes Prediction Dataset." Kaggle, 2023, www.kaggle.com/datasets/iammustafatz/diabetes-prediction-dataset.

Questions?